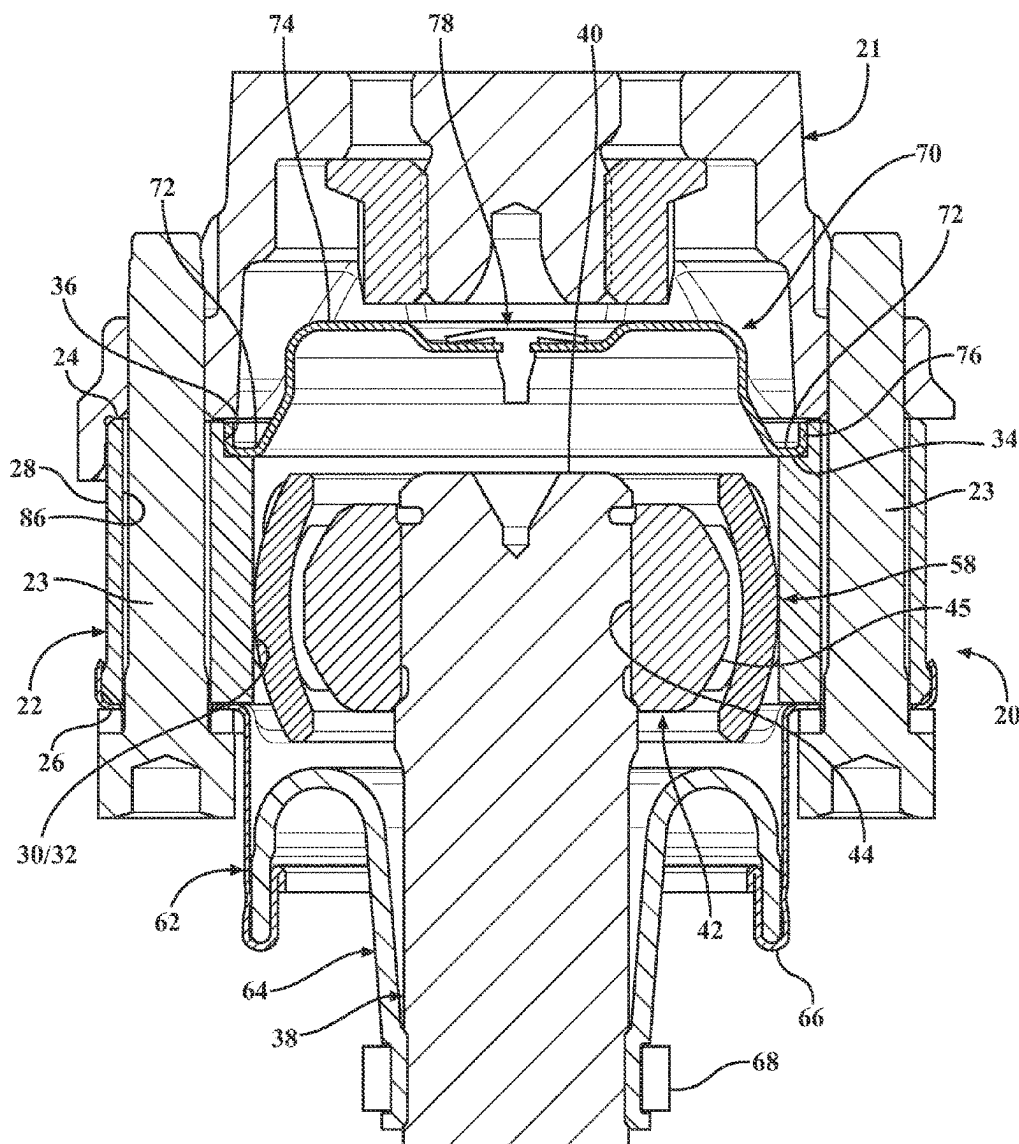




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(19) **United States**(12) **Patent Application Publication**
HORNE et al.(10) **Pub. No.: US 2025/0271037 A1**(43) **Pub. Date: Aug. 28, 2025**(54) **CONSTANT VELOCITY JOINT**(52) **U.S. Cl.**CPC **F16D 3/226** (2013.01); **F16D 2003/22326**
(2013.01)(71) Applicant: **Neapco Intellectual Property**
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(US)(57) **ABSTRACT**(72) Inventors: **Aaron HORNE**, Lathrup Village, MI
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A constant velocity joint includes an outer race that extends about an axis and is configured to be coupled with a first shaft. The outer race extends axially between a top surface and a bottom surface. An inner race is located in the compartment and is pivotable relative to the outer race for being connected with a second shaft. A plurality of balls are positioned between the outer and inner races for guiding the pivoting movement outer and inner races while transmitting rotational movement between the outer and inner races. An upper can is coupled to the top surface of the outer race for sealing the compartment of the outer race. A plurality of bolt holes extend axially through the outer race radially outwardly of the upper can.

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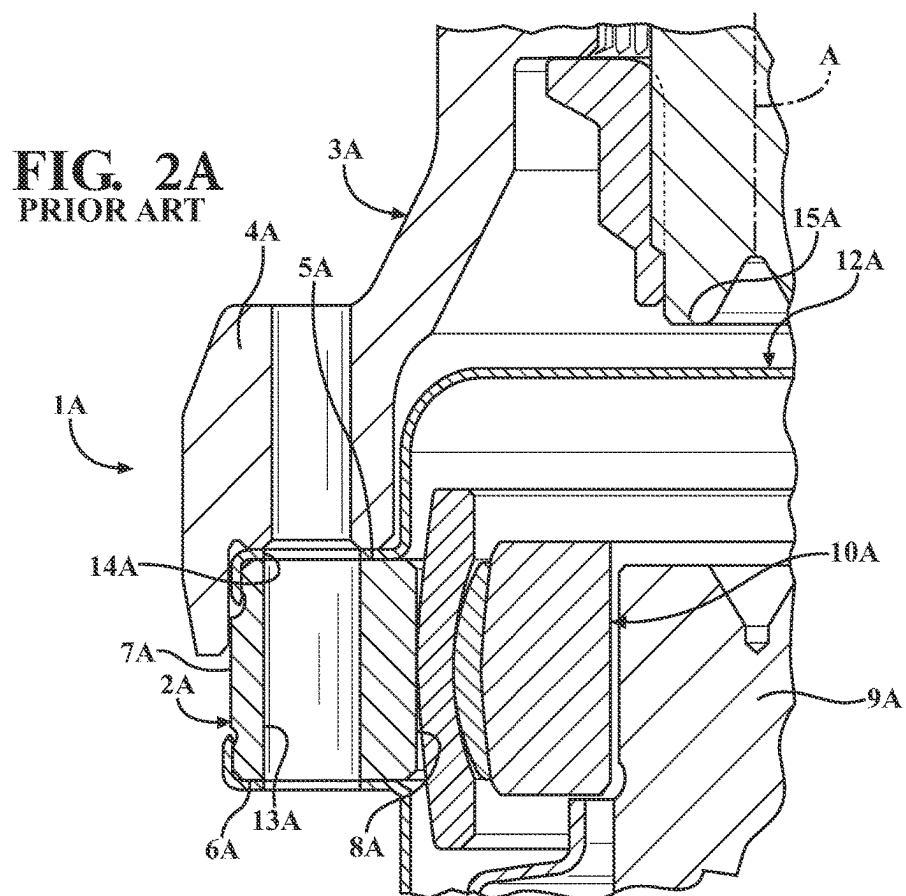
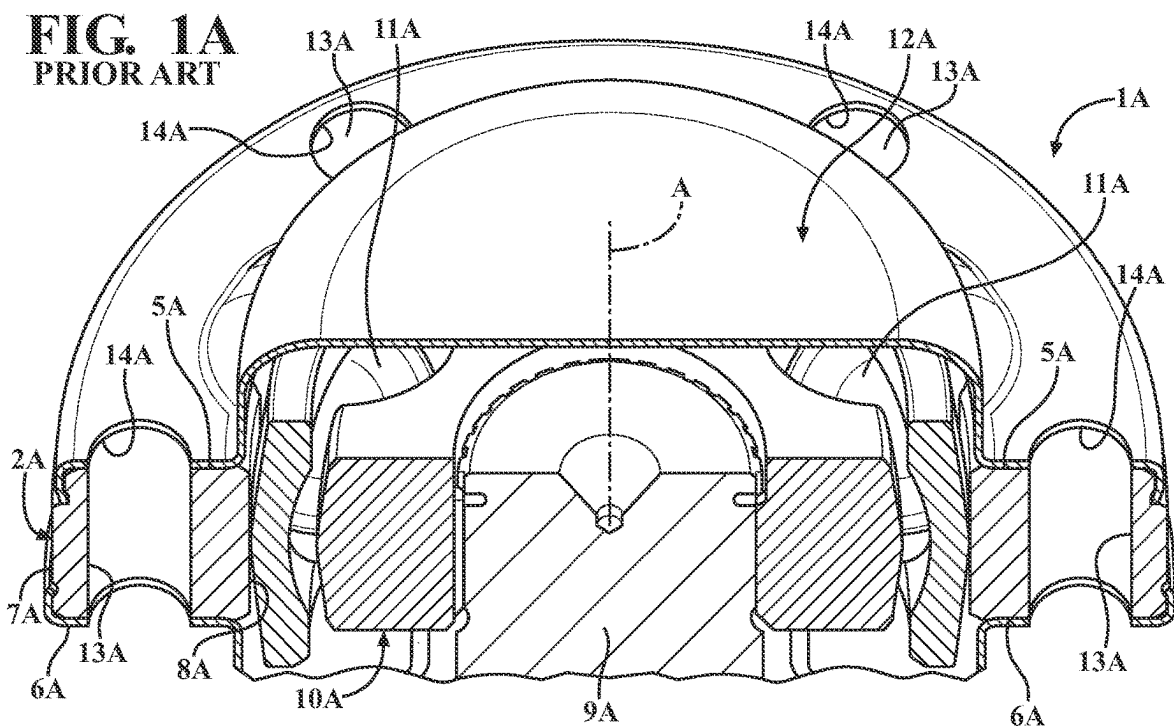


FIG. 1B
PRIOR ART

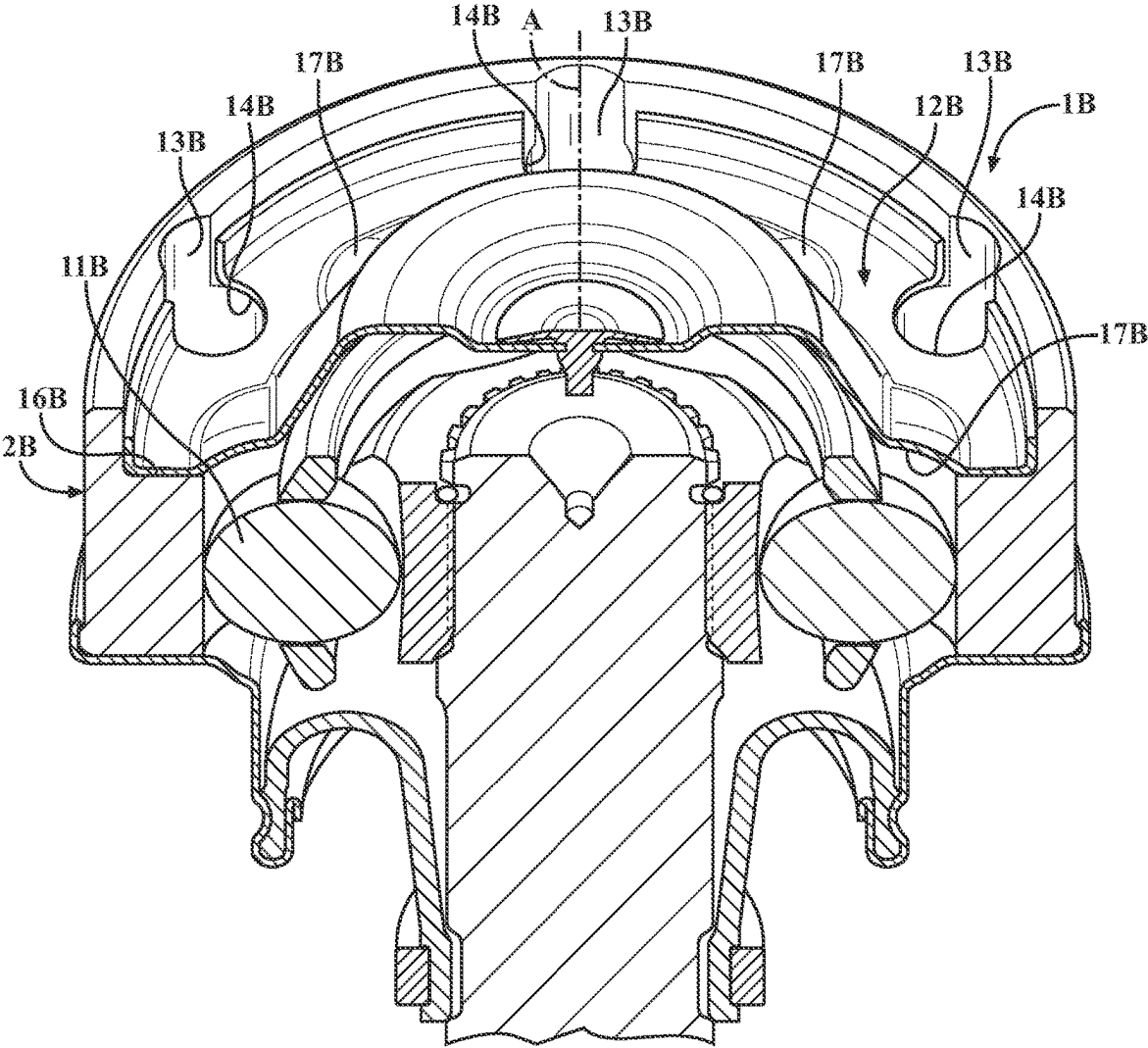
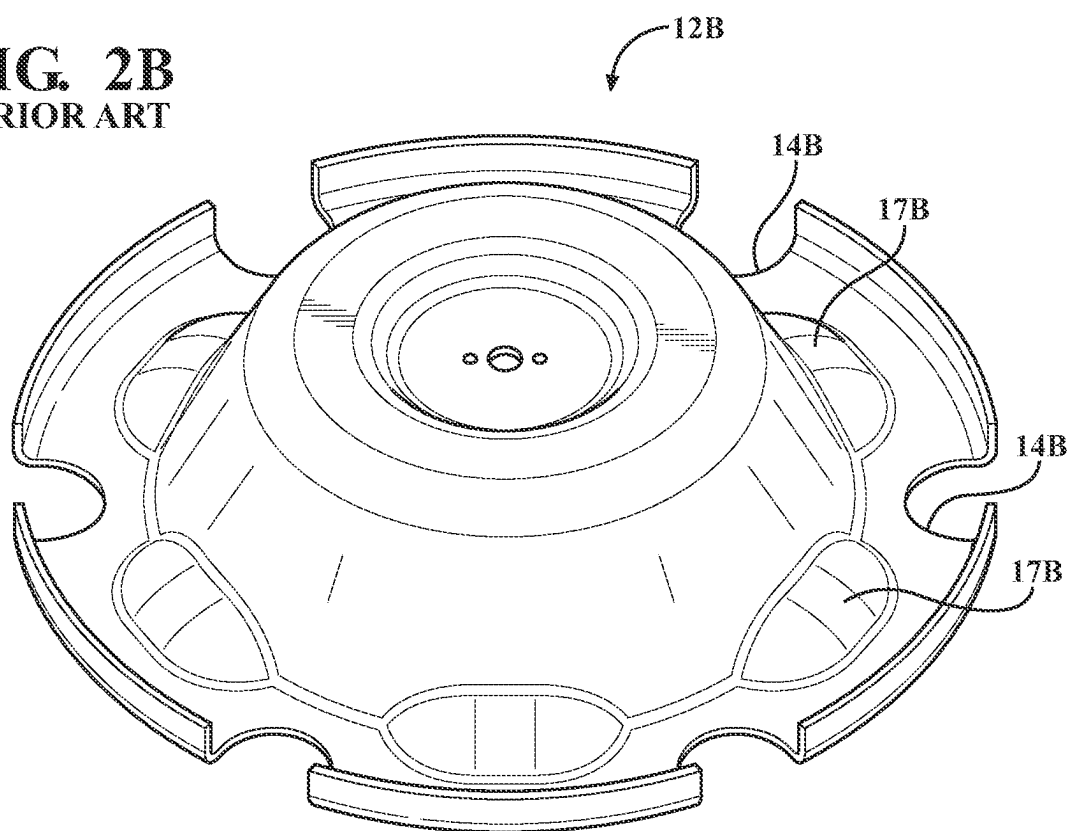


FIG. 2B
PRIOR ART



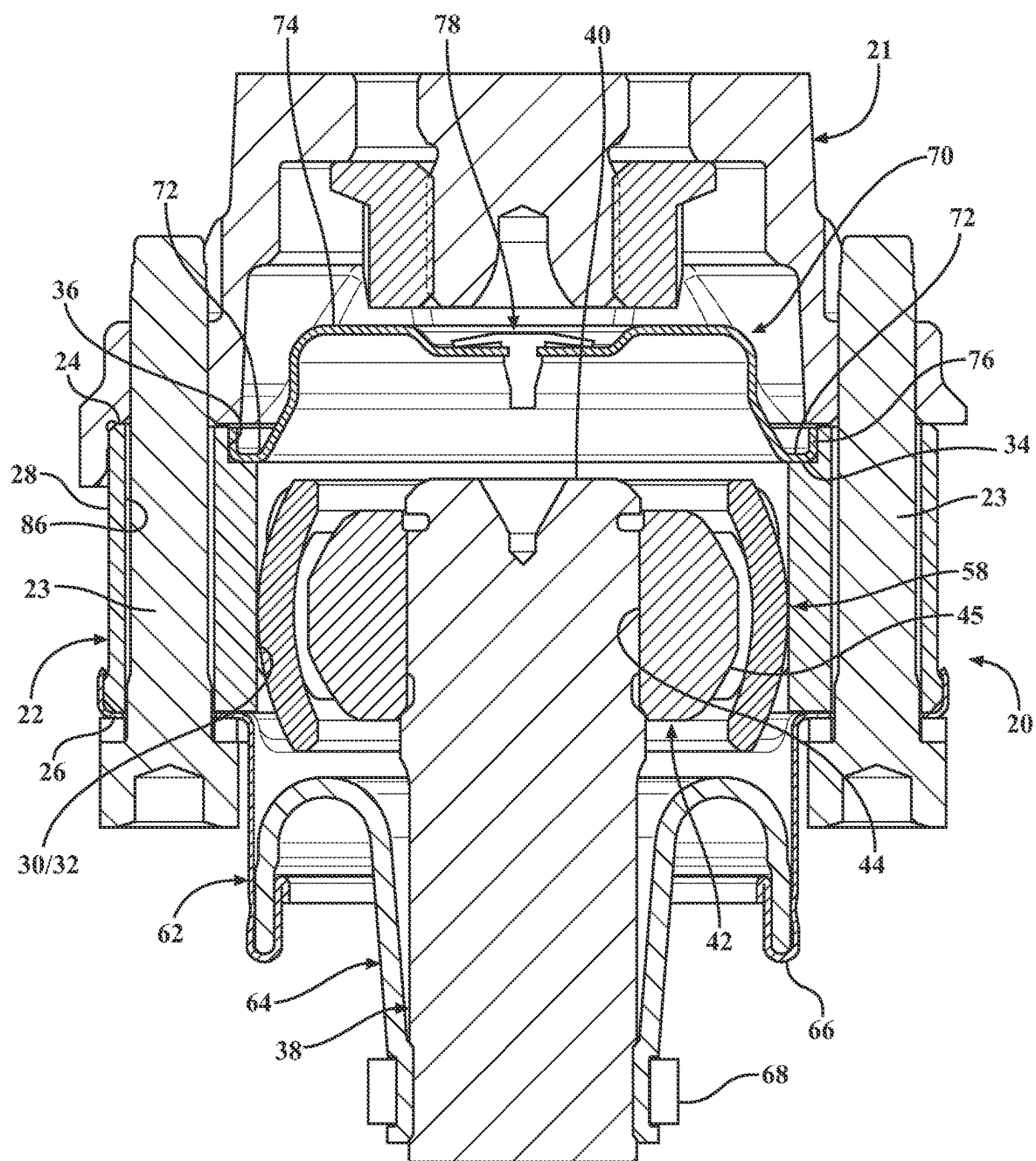
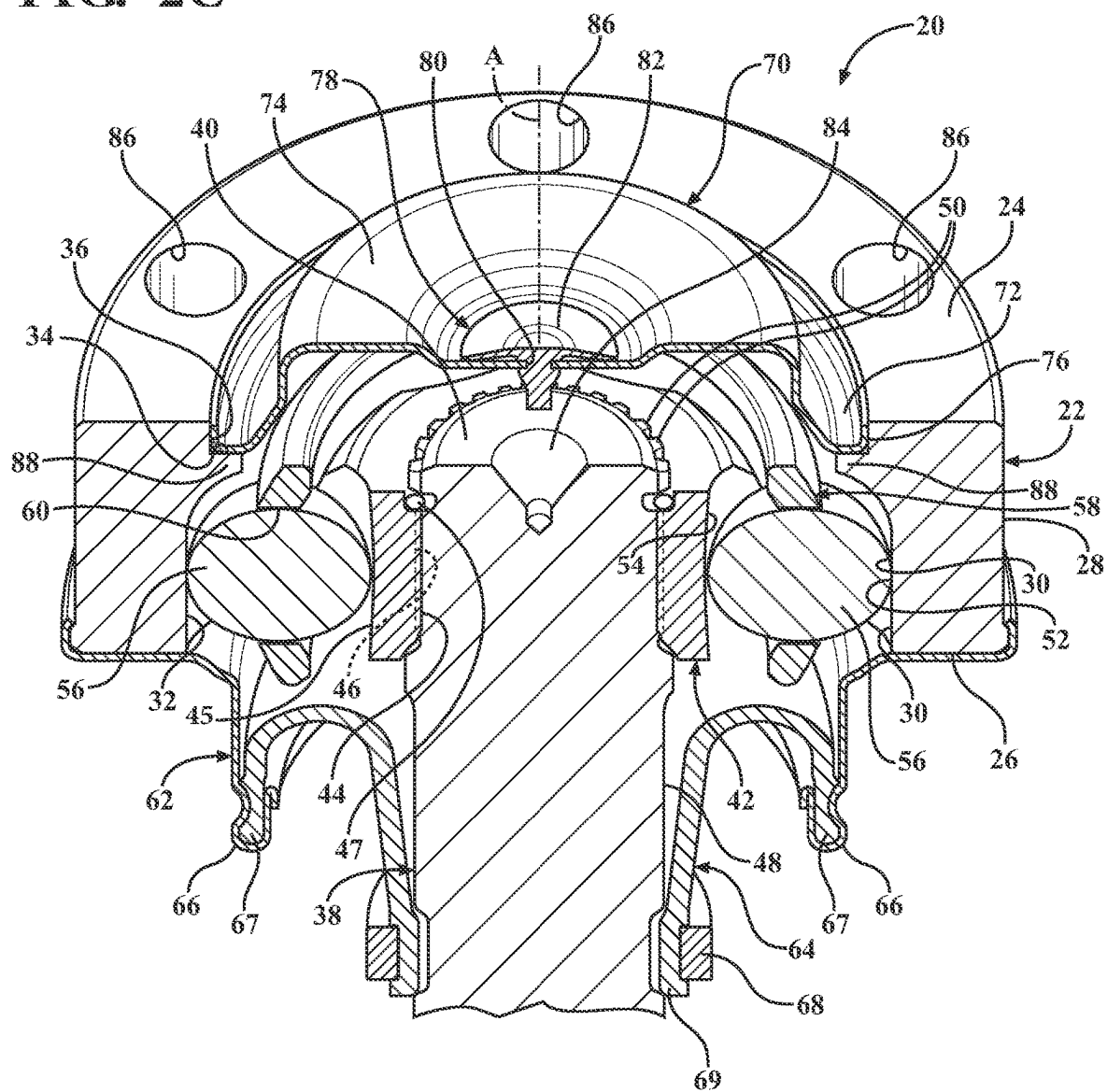


FIG. 1C

FIG. 2C



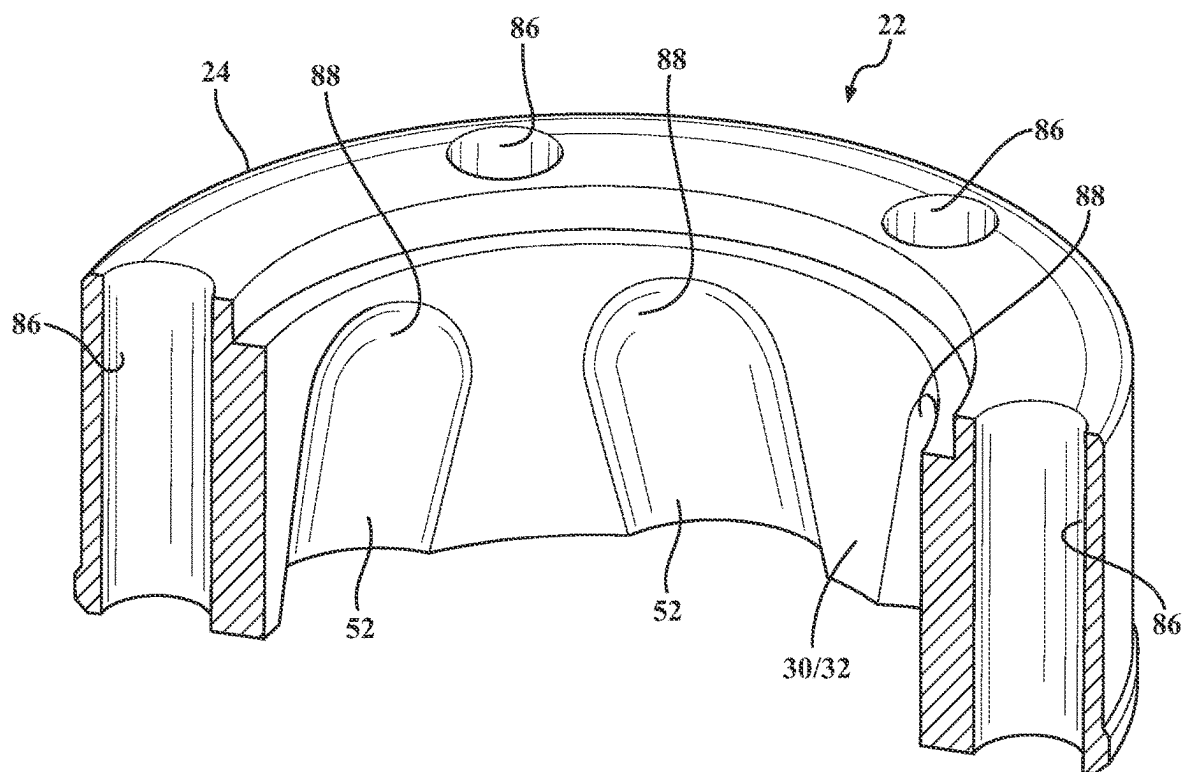


FIG. 3C

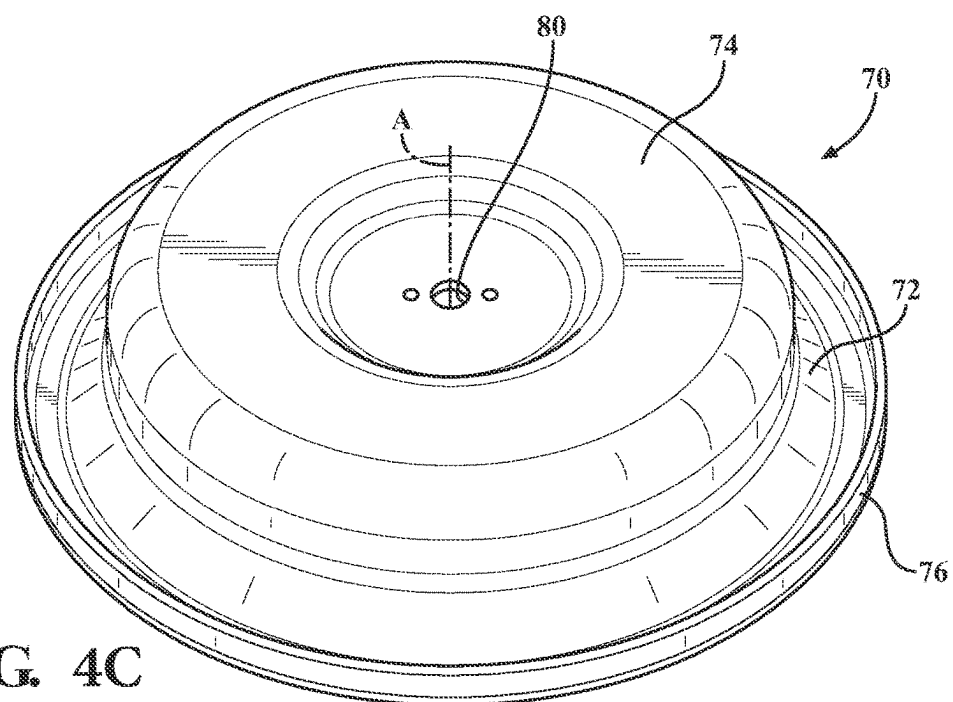


FIG. 4C

CONSTANT VELOCITY JOINT

FIELD

[0001] The present disclosure relates generally to vehicle driveline components. More particularly, the present disclosure relates to constant velocity joints for transmitting rotational movements between two shafts while permitting relative angular deflections between the two shafts.

BACKGROUND

[0002] This section of the written disclosure provides background information related to constant velocity joints and is not necessarily prior art to the inventive concepts disclosed and claimed in this application.

[0003] Constant velocity joints (CV joints) are used on drivelines of vehicles for transmitting rotational movement between two shafts while permitting angular deflections of the shafts relative to one another. An example of a prior art constant velocity joint is shown in FIGS. 1A to 2A. The prior art constant velocity joint 1A includes an outer race 2A that extends about an axis A and is configured to be coupled with a first shaft 3A (shown in FIG. 2A). The first shaft 3A includes a flange portion 4A that is configured to be positioned about and above the outer race 2A and coupled to the outer race 2A via bolts (not shown). The outer race 2A extends axially between a top surface 5A and a bottom surface 6A, and extends radially between a radially outer surface 7A and a radially inner surface 8A, which defines a compartment. A second shaft 9A extends along the axis A to a terminal end in the compartment of the outer race 2A. An inner race 10A is coupled with the second shaft 9A in the compartment and is pivotable relative to the outer race 2A. A plurality of balls 11A are positioned radially between the outer and inner races 2A, 10A for guiding the pivoting movement of the inner race 10A relative to the outer race 2A while transmitting rotational movement between the outer and inner races 2A, 10A. A press-fitted upper can 12A (cover) is coupled to the top surface 5A of the outer race 2A for sealing the compartment of the outer race 2A. As shown, the upper can 12A is formed over the outer race 12A along the radially outer surface 7A. A plurality of bolt holes 13A extend axially through the outer race 2A and are arranged circumferentially around the outer race 2A. Likewise, the upper can 12A defines a plurality of bolt holes 14A, each in alignment with one of the bolt holes 13A of the outer race 2A. A plurality of bolts (not shown) pass through the aligned bolt holes 13A, 14A to secure the outer race 2A and first shaft 3A to one another, with the upper can 12A serving as a sealing mechanism. This arrangement is typically used when there is ample storage space to accommodate the flange portion 4A of the first shaft 3A, and when there is a satisfactory axial gap between the upper can 12A and a flange nut 15A of the first shaft 3A (best shown in FIG. 2A). In cases in which there are restricted space requirements for the CV joint 1A, this arrangement can be unfeasible. More particularly, an inside diameter of the flange portion 4A of the first shaft 3A may not be large enough to support a required bolt circle diameter of the bolts along with the externally press fit upper can 12A without resulting in an insufficient cross-sectional thickness around the bolt holes 13A, 14A.

[0004] Thus, an alternative prior art CV joint 1B shown in FIGS. 1B and 2B includes a “recessed rear cover/can” type

design to facilitate use of the CV joint 1B in restricted space applications. According to this design, instead of being clamped to an outside diameter of the outer race 2B, an upper can 12B is pressed into a counter bore 16B on the top surface 5B of the outer race 2B. This arrangement has a small diametrical profile while also providing ample axial clearance to an axle nut of a first shaft. The upper can 12B of this arrangement includes scallops (cutouts) 14B aligned with the bolt holes 13B of the outer race 2B, as well as a series of circumferentially spaced pockets 17B aligned with the balls 11B for inhibiting axial movement of the balls 11B. The upper can 12B of this design is secured to the outer race 2B via a press fit in the counter bore 16B. While this arrangement is more compact than the CV joint 1A of FIGS. 1A and 2A, the scallops 14B and pockets 17B of the upper can 12B can result in interrupted sealing along press fit areas against the outer race 2B, which can cause the CV joint 1B to be susceptible to grease loss and spray of the grease against an underside of the associated vehicle during use. More particularly, the scallops 14B and pockets 17B leave a relative small surface area for sealing the upper can 12B to the outer race 2B for retaining grease within the outer race 2B. The scallops 14B and ball pockets 17B are also required to have tight tolerances and must be precisely positioned relative to the outer race 2B during installation to provide a good seal. It is known to use sealants such as RTV (room temperature vulcanizing) at the interface between the outer race 2B and the upper can 12B to reduce risk of grease loss and ingress of external contaminants, but the use of such sealants may inadequately prevent grease leakage.

[0005] In view of the foregoing, there remains a need for further improvements to CV joint assemblies, particularly those intended for use in restricted space applications.

SUMMARY

[0006] This section provides a general summary of the inventive concepts associated with this disclosure and is not intended to be interpreted as a complete and comprehensive listing of all of its aspects, objectives, features and advantages.

[0007] According to an aspect of the disclosure, a constant velocity joint includes an outer race that extends about an axis and is configured to be coupled with a first shaft. The outer race extends axially between a top surface and a bottom surface, and extends radially between a radially outer surface and a radially inner surface. The radially inner surface of the outer race defines a compartment. An inner race is located in the compartment for being coupled with a second shaft and is pivotable relative to the outer race. A plurality of balls are positioned radially between the outer and inner races for guiding the pivoting movement of the inner race relative to the outer race while transmitting rotational movement between the outer and inner races. An upper can is coupled to the top surface of the outer race for sealing the compartment of the outer race. A plurality of bolt holes extend axially through the outer race and are arranged circumferentially around the outer race and positioned radially outwardly of the upper can for receiving bolts to attach the outer race to the first shaft.

[0008] The position of the upper can radially inwardly of the bolt holes eliminates the need for the bolts to pass through the upper can, which provides a large sealing area of the upper can to the outer race 360 degrees circumferentially around the entire upper can. This minimizes grease

loss and spray to the underside of the vehicle. Furthermore, the upper can is simple in design and easy to fabricate in that can to be made via a simple stamping process.

[0009] According to another aspect of the disclosure, a constant velocity joint includes an outer race that extends about an axis and is configured to be coupled with a first shaft. The outer race extends axially between a top surface and a bottom surface and extends radially between a radially outer surface and a radially inner surface. The radially inner surface of the outer race defines a compartment. An inner race is located in the compartment for being coupled with the second shaft in the compartment and is pivotable relative to the outer race. A plurality of balls are positioned radially between the outer and inner races for guiding the pivoting movement of the inner race relative to the outer race while transmitting rotational movement between the outer and inner races. An upper can is coupled to the top surface of the outer race for sealing the compartment of the outer race. The inner circumference of the outer race presents at least one rim that extends radially over the balls under the counter bore of the outer race to define at least one pocket that receives the balls to limit axial movement of the balls.

[0010] Locating the at least one rim on the outer race eliminates the need to form ball pockets on the upper can, thus eliminating required design/manufacturing tolerances and permitting the upper can to be fabricated in a simple manner. Also, because the rim is located beneath the top surface of the outer race instead of on the upper can, space is provided at the top surface of the outer race to form a counter bore for holding the upper can radially inwardly of the bolt holes.

BRIEF DESCRIPTION OF DRAWINGS

[0011] The drawings described herein are for illustrative purposes only of selected embodiments and not all possible implementations, and are not intended to limit the scope of the present disclosure.

[0012] FIG. 1A is a perspective cross-sectional view of a first conventional arrangement of a constant velocity joint;

[0013] FIG. 2A is a side, cross-sectional view of a connection between the first conventional arrangement of a constant velocity joint and a first shaft;

[0014] FIG. 1B is a perspective cross-sectional view of a second conventional arrangement of a constant velocity joint;

[0015] FIG. 2B is a perspective view of an upper can of the second conventional arrangement of a constant velocity joint;

[0016] FIG. 1C is a side, cross-sectional view of an embodiment of a constant velocity joint according to an aspect of the present disclosure, illustrating a connection to first and second shafts;

[0017] FIG. 2C is a perspective, cross-sectional view of the constant velocity joint of FIG. 1C;

[0018] FIG. 3C is a perspective view of an upper can of the constant velocity joint of FIG. 1C; and

[0019] FIG. 4C is a perspective, cross-sectional view of an outer race of the constant velocity joint of FIG. 1C, illustrating an arrangement of first ball tracks and associated rims for limiting movement of balls.

DETAILED DESCRIPTION

[0020] Exemplary embodiments will now be described more fully with reference to the accompanying drawings. In particular, a number of non-limiting embodiments of a constant velocity joint **20** (“CV joint”) are provided so that this disclosure will be thorough and will fully convey the true and intended scope to those who are skilled in the art. It will be apparent to those skilled in the art that specific details need not be employed, that example embodiments may be embodied in many different forms and that neither should be construed to limit the scope of the disclosure. In some example embodiments, well-known processes, well-known device structures, and well-known technologies are not described in detail. It should also be appreciated that the present invention can be utilized in connection with other types of automobile components not described fully herein.

[0021] The CV joint **20** could be used on various vehicle components such as propeller shafts, drive shafts, half shafts, axles, transfer cases, power takeoff (“PTO”) units, and/or other components that operate to transmit rotational forces (i.e., rotational energy), between and/or through one or more other driveline components.

[0022] With reference to FIGS. 1C-4C, the CV joint **20** includes an outer race **22** that extends about an axis A and is configured to be coupled with a first shaft **21** via an arrangement of a number of bolts **23**. The outer race **22** extends axially between a top surface **24** and a bottom surface **26**, and extends radially between a radially outer surface **28** and a radially inner surface **30**. The radially inner surface **30** defines a compartment **32**. The top surface **24** of the outer race **22** defines a counter bore **34, 36** that extends axially downwardly along the radially inner surface **30**. More particularly, the counter bore **34, 36** is defined by an annular radial wall **34** that extends along a plane that is perpendicular to the axis A, and an annular axial wall **36** that extends axially. As will be described in further detail below, the top surface **24** of the outer race **22** defines a plurality of bolt holes **86** that are spaced circumferentially from one another around the outer race **22** and around the counter bore **34, 36** for connecting the outer race **22** to the first shaft **21** via the arrangement of bolts **23**.

[0023] A second shaft **38** extends along the axis A to a terminal end **40** in the compartment **32** of the outer race **22**. An inner race **42** is coupled with the second shaft **38** adjacent to the terminal end **40** in the compartment **32**. The inner race **42** is pivotable relative to the outer race **22** such that the second shaft **38** and inner race **42** together are pivotable relative to the outer race **22** and the first shaft. The inner race **42** has a radially inward surface **44** and a radially outward surface **45**.

[0024] As best shown in FIG. 2C, the radially inward surface **44** of the inner race **42** defines a plurality of axial slots **46** that are arranged in circumferentially spaced relationship with one another around the radially inward surface **44**. An outside surface **48** of the second shaft **38** presents a plurality of axially extending ribs **50** positioned in circumferentially spaced relationship with one another around the outside surface **48**. Each of the ribs **50** are received by one of the axial slots **46** of the inner race **42** to rotationally connect the second shaft **38** and the inner race **42** while accommodating axial movement of the second shaft **38** relative to the inner race **42** to connect the second shaft **38** to the inner race **42**. Furthermore, a retaining ring **47**, is

positioned radially between the second shaft **38** and the inner race **42** to affix the inner race **42** to the second shaft **38**.

[0025] The radially inner surface **30** of the outer race **22** defines a plurality of first ball tracks **52** that are arranged in circumferentially spaced relationship with one another. The radially outward surface **45** of the inner race **42** defines a plurality of second ball tracks **54**, each in circumferential alignment with the first ball tracks **52** of the outer race **22**. A plurality of balls **56** are positioned radially between the outer and inner races **22**, **42**, each in one of the first and second ball tracks **52**, **54** for guiding the pivoting movement of the outer and inner races **22**, **42** relative to one another while also transmitting rotational movement between the outer and inner races **22**, **42**. As best illustrated in FIG. 3C, the radially inner surface **30** of the outer race **22** presents a plurality of rims **88** at an axial top of each of the first ball tracks **52**. Each of the rims **88** are hemispherical-shaped and extend radially over the balls **56** axially under the counter bore **34**, **36** of the outer race **22** to define a pocket for axially limiting movement of/containing the balls **56**. Because the first ball tracks **52** are truncated in this manner beneath the top surface **24** of the outer race **22** to form the rims **88**, it is feasible to form the counter bore **34**, **36** radially inwardly of the bolt holes **86**. This is contrary to prior CV joints in which pockets are formed in an upper can.

[0026] With reference back to FIGS. 1C-2C, a cage **58** is positioned in the compartment **32** between the outer and inner races **22**, **42**. The cage **58** defines a plurality of openings **60** each for receiving one of the balls **56** for aligning the balls **56** in predetermined locations.

[0027] A lower can **62** is fixed to the bottom surface **26** of the outer race **22** about the second shaft **38**. A boot seal **64** extends radially between, and is sealed against the second shaft **38** and the lower can **62**. The lower can **62** has a crimped end **66** that is crimped about a first end **67** of the boot seal **64** in order to seal the first end **67** of the boot seal **64** to the lower can **62**. Furthermore, a clamp **68** is wrapped annularly about a second end **69** of the boot seal **64** and the second shaft **38** to seal the second end **69** of the boot seal **64** to the second shaft **38**.

[0028] As best shown in FIGS. 2C and 4C, a generally cup-shaped upper can **70** is coupled with the top surface **24** of the outer race **22** in the counter bore **34**, **36** of the outer race **22**. The upper can **70** has a rim portion **72** and a protruding portion **74** located centrally relative to the rim portion **72**. The upper can **70** extends axially upwardly relative to the rim portion **72**. The rim portion **72** overlies the radial wall **34** of the counter bore **34**, **36** of the outer race **22** and terminates at a lip **76** that extends axially upwardly and engages the axial wall **36** of the counter bore **34**, **36**. The upper can **70** is fitted within the counter bore **34**, **36** with a press/interference fit. Together, the lower and upper cans **62**, **70** and the boot seal **64** are configured to seal the compartment **32** to prevent contaminants from entering the compartment **32**.

[0029] The upper can **70** defines a vent plate portion **78** along the axis A. The vent plate portion **78** includes a mounting aperture **80** that is defined by the upper can **70** along the axis A. A vent valve **82** is received by the mounting aperture **80** for sealing the compartment **32** and to operate as a one-way valve to allow air to escape the compartment **32** when the air in the compartment **32** reaches a predetermined pressure, while preventing contaminants from entering the compartment **32**. The terminal end **40** of the second shaft **38**

may define a recess **84** for aiding in positioning the second shaft **38** during various manufacturing operations.

[0030] As previously noted, the top surface **24** of the outer race **22** defines the plurality of bolt holes **86**. The bolt holes **86** extend axially through the outer race **22** and are arranged circumferentially around the outer race **22**. The bolt holes **86** are positioned radially outwardly of the counter bore **34**, **36** and the upper can **70** for receiving bolts **23** to attach the outer race **22** to the first shaft (best shown in FIG. 1C).

[0031] Because the upper can **70** is located radially inwardly of the bolt holes **86**, it provides continuous and uninterrupted sealing 360 degrees around the upper can **70** against outer race **22**. This is contrary to prior art CV joints which had scallops for receiving bolts defined in the upper can which were prone to inconsistent sealing and grease leaks. Furthermore, the arrangement of the upper can **70** being free of ball pockets due to the presence of the rims **88** of the outer race **22** removes design/manufacturing tolerances and provides simpler positioning of the upper can **70**.

[0032] Obviously, many modifications and variations of the present invention are possible in light of the above teachings and may be practiced otherwise than as specifically described while within the scope of the appended claims. Individual elements or features of a particular embodiment are generally not limited to that particular embodiment, but, where applicable, are interchangeable and can be used in a selected embodiment, even if not specifically shown or described.

What is claimed is:

1. A constant velocity joint, comprising:

an outer race extending about an axis and configured to be coupled with a first shaft;

the outer race extending axially between a top surface and a bottom surface and extending radially between a radially outer surface and a radially inner surface, the radially inner surface of the outer race defining a compartment;

an inner race in the compartment and pivotable relative to the outer race for being connected to a second shaft;

a plurality of balls positioned radially between the outer and inner races for guiding the pivoting movement of the inner race relative to the outer race while transmitting rotational movement between the outer and inner races;

an upper can coupled to the top surface of the outer race for sealing the compartment of the outer race; and

a plurality of bolt holes extending axially through the outer race and arranged circumferentially around the outer race and positioned radially outwardly of the upper can for receiving bolts to attach the outer race to the first shaft.

2. The constant velocity joint as set forth in claim 1, wherein the top surface of the outer race defines a counter bore along the radially inner surface of the outer race radially inward of the bolt holes, and wherein the upper can is fixed in the counter bore.

3. The constant velocity joint as set forth in claim 2, wherein the upper can is press fit in the counter bore.

4. The constant velocity joint as set forth in claim 2, wherein the upper can has a rim portion in the counter bore, and wherein the rim portion engages the top surface of the outer race in the counter bore continuously and uninterrupted 360 degrees around the rim.

5. The constant velocity joint as set forth in claim 4, wherein the counter bore includes a radial wall extending along a plane being perpendicular to the axis and a axial wall extending axially, and wherein the rim portion overlies the radial wall of the counter bore of the outer race and terminating at a lip extending axially upwardly and engaging the axial wall of the counter bore.

6. The constant velocity joint as set forth in claim 1, wherein the inner circumference of the outer race presents at least one rim extending radially over the balls under the counter bore of the outer race to limit axial movement of the balls.

7. The constant velocity joint as set forth in claim 6, wherein the at least one rim is hemispherical shaped.

8. The constant velocity joint as set forth in claim 6, wherein the radially inner surface of the outer race defines a plurality of axially extending first ball tracks, a radially outward surface of the inner race defines a plurality of axially extending second ball tracks, wherein the first and second ball tracks are circumferentially aligned with one another with one of the plurality of balls positioned between each of the aligned ball tracks, and wherein the at least one rim includes a plurality of rims each located at a top of one of the first ball tracks of the outer race.

9. A constant velocity joint, comprising:

an outer race extending about an axis and configured to be coupled with a first shaft;

the outer race extending axially between a top surface and a bottom surface and extending radially between a radially outer surface and a radially inner surface, the radially inner surface of the outer race defining a compartment;

an inner race located in the compartment and pivotable relative to the outer race for being connected with a second shaft;

a plurality of balls positioned radially between the outer and inner races for guiding the pivoting movement of the inner race relative to the outer race while transmitting rotational movement between the outer and inner races;

an upper can coupled to the top surface of the outer race for sealing the compartment of the outer race; and

the inner circumference of the outer race presenting at least one rim and extending radially over the balls under the counter bore of the outer race to inhibit axial movement of the balls.

10. The constant velocity joint as set forth in claim 9, wherein the rim is hemispherical shaped.

11. The constant velocity joint as set forth in claim 10, wherein the radially inner surface of the outer race defines a plurality of axially extending first ball tracks, a radially outward surface of the inner race defines a plurality of axially extending second ball tracks, wherein the first and second ball tracks are circumferentially aligned with one another and with one of the plurality of balls positioned between each of the aligned ball tracks, and wherein the at least one rim includes a plurality of rims each located at a top of one of the first ball tracks of the outer race.

12. The constant velocity joint as set forth in claim 10, wherein the top surface of the outer race defines a plurality of bolt holes extending axially through the outer race and arranged circumferentially around the outer race and positioned radially outwardly of the upper can for receiving bolts to attach the outer race to the first shaft.

13. The constant velocity joint as set forth in claim 10, wherein the top surface of the outer race defines a counter bore along the radially inner surface of the outer race radially inward of the bolt holes, and wherein the upper can is fixed in the counter bore.

14. The constant velocity joint as set forth in claim 13, wherein the upper can has a rim portion in the counter bore, and wherein the rim portion engages the top surface of the outer race in the counter bore continuously and uninterrupted 360 degrees around the rim.

15. The constant velocity joint as set forth in claim 13, wherein the upper can is press fit in the counter bore.

16. The constant velocity joint as set forth in claim 13, wherein the counter bore includes a radial wall extending along a plane being perpendicular to the axis and a axial wall extending axially, and wherein the upper can has a rim portion overlying the radial wall of the counter bore of the outer race and terminating at a lip extending axially upwardly and engaging the axial wall of the counter bore.

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